MongoDB Practice Task (Products & Orders Collections)

Question 1

Find all products that belong to the Electronics category and have a price greater than ₹5000.

```
db.products.find({
  category:"Electronics",
  price:{$gt:5000}
})
```

Question 2

Display each order with the Product Name, Price, and Quantity.

```
db.orders.aggregate([
  {
    $lookup:{
      from:"products",
      localField:"product_id",
      foreignField:"_id",
      as:"product_details"
    }
  },
  {
    unwind: "product_details"
  },
```

```
  {
    $project:{
      _id:0,
      productName:"$product_details.name",
      price:"$product_details.price",
      quantity:1
    }
  }
])
```

Question 3

Find all products whose stock is between 20 and 80 (inclusive).

```
db.products.find({
  stock:{
    $gt:20,
    $lt:80
  }
})
```

Question 4

Calculate the average price of products in each category. Display only:

- **Category**
- **Average Price**

Sort the result by average price in descending order.

```
db.products.aggregate([
```

```
  {
    $group:{
      _id:"$category",
      averagePrice:{$avg: "price"}
    }
  },
  {
    $project:{
      category:"$_id",
      averagePrice:1
    }
  },
  {
    $sort:{
      averagePrice:-1
    }
  }
])
```

Question 5

Find all products that are not in the Electronics category.

```
db.products.find({
  category:{$ne:"Electronics"}
})
```

Question 6

Display each order with:

- **Product Name**
- **Category**
- **Price**
- **Quantity**

Sort the results by product price in descending order.

```
db.orders.aggregate([
  {
    $lookup:{
      from:"products",
      localField:"product_id",
      foreignField:"_id",
      as:"productDetails"
    }
  },
  {
    unwind: "productDetails"
  },
  {
    $project:{
      _id:0,
      productName:"$productDetails.name",
      category:"productDetails.category", price: "productDetails.price",
```

```
      quantity:1
    }
  },
  {   $sort:{      price:-1   }
}
])
```

Question 7

Find all products whose price is less than ₹3000 and stock is greater than 20.

```
db.products.find({
 $and:[{
   price:{$lt:3000},
   stock:{$gt:20}
 }]
})
```

Question 8

Calculate the total inventory value for each product.

Formula:

Total Value = Price × Stock

Display:

- **Product Name**
- **Category**

- Total Inventory Value

Sort by Total Inventory Value in descending order.

```
db.products.aggregate([
  {
    $project:{
      _id:0,
      productName:"$name",
      category:1,
      totalInventryValue:{
        multiply: ["price","$stock"]
      }
    }
  },
  {
    $sort:{totalInventryValue:1}
  }
])
```

Question 9

Find all products whose category is either Electronics or Clothing, and whose price is greater than ₹1000.

```
db.products.find({
  category:{
```

```
    $in:["Electronics","Clothing"]

  },

  price:{$gt:2000}

})
```

Question 10

Calculate the total amount for every order.

Formula:

Total Amount = Product Price × Quantity

Display:

- **Product Name**
- **Quantity**
- **Price**
- **Total Amount**

Sort by Total Amount in descending order.

```
db.orders.aggregate([

  {

    $lookup:{

      from:"products",

      localField:"product_id",

      foreignField:"_id",

      as:"product_details"

    }

  },

  {
```

```
    unwind: "product_details"
  },
  {
    $project:{
      _id:0,
      productName:"$product_details.name",
      quantity:1,
      price:"$product_details.price",
      totalAmount:{
        multiply: ["product_details.price","$quantity"]
      }
    }
  },
  {
    $sort:{totalAmount:-1}
  }
])
```